Transient Stability Analysis of the IEEE 14-Bus Electric Power System

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Abstract- **It is widely accepted that transient stability is an important aspect in designing and upgrading electric power system. This paper covers the modeling and the transient stability analysis of the IEEE 14 test bus system using Matlab Power System Toolbox (PST) package. A three-phase fault is located at two different locations, to analyze the effect of fault location and critical clearing time on the system stability. In order to protect overhead transmission line, conductors and insulators, it is suggested that the faulted part to be isolated rapidly from the rest of the system so as to increase stability margin and hence decrease damage.**

Keywords: Critical Clearing Time, Power System Stability, Synchronous Generation and Transient Stability.

1. Introduction

As electric utilities have grown in size, and the numbers of interconnections have increased, planning for future expansion has become increasingly complex. The increasing cost of addition and modifications has made it imperative that utilities consider a range of design options, and perform detailed studies of the effect on the system of each option, based on a number of assumptions: normal and abnormal operating conditions, peak and off-peak loadings, and present and future years of operation. One of the important studies that must be included in designing an electric power system is "Transient Stability".

Transient stability entails the evaluation of a power system's ability to withstand large disturbances, and to survive transition to a normal operating condition. These disturbances can be faults such as: a short circuit on a transmission line, loss of a generator, loss of a load, gain of load or loss of a portion of transmission network [2]. Large number of simulations is carried out regularly during planning stages to gain knowledge of this system. Yet, even a well designed and normally operated system may face the threat of transient instability.

The aim of the investigation is to analyze the behavior of the synchronous machine in particular the angular position of the rotor with respect to time after the fault occurs in the system. Section 2 describes the development of the IEEE 14-bus system model. The case study of IEEE-14 bus system and the results from the case study are presented in section 3 and Section 4 respectively. Section 5 concludes the paper.

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2. Development of a Model of the IEEE 14 Bus System

In the stability study for the considered IEEE 14-bus system, the following assumptions are accepted [3]: (i) the input remains constant during the entire period of a swing curve; (ii) damping or synchronous power is proportional to the generator's own relative speed or slip; (iii) synchronous power is calculated from a steady-state solution of the network; (iv) each machine is represented in the network by a constant reactance (direct-axis transient reactance) in series with the constant electromotive force (the internal voltage); (v) the mechanical angle *X(i)* of each machine rotor coincides with the electrical phase of the internal voltage; (vi) the load is represented by shunt admittances to its buses. There are three states in the system:

1) The prefault state which determines the initial condition for angles *X(i)* where *i*= 1,2,3,4,5 (equals to the number of synchronous machine in the system)
2) The fault state which exists at *t*=0 and persists until the fault is cleared at *t=tcr;*
3) The postfault state with *t>tcr.*

where *t[s]* is fault clearing time and *tcr[s]* is critical clearing time.

2.1 Basic Swing Equation for the power system

When performing a stability study, the necessary starting point is to determine the description of the differential equations for the system. For a typical generator system, the mechanical input torque is given by [4]

$$T_m = J\alpha_m + T_D + T_e \tag{1}$$

where T_m *[Nm]* is the mechanical energy input at the rotor shaft . J *[kg m^2]* is the combined polar moment of the inertia of the rotor masses , α_m *[rad/sec^2]* is the acceleration of the rotor masses, T_D *[Nm]* is the damping torque of the generator squiller cage winding and T_e [*Nm*] is the torque equivalent of the generator electrical output power. The damping term T_D in equation (1) is a very small percentage of T_e and thus equation (1) is sometimes approximated by [2]

$$T_m = J\alpha_m + T_e \qquad (2)$$

This equation can be represented in terms of electrical power as [5]

$$\frac{d^2X}{dt^2} = \frac{\pi f}{H}(P_m - P_e) \qquad (3)$$

Where X is the rotor angle of the machine with respect to the synchronously rotating reference frame in radians and

$$H = \frac{\text{stored rotor kinetic energy at synchronou s speed } [MJ]}{\text{generator rating } [MVA]} \qquad (4)$$

f is the system frequency, P_m is the mechanical input power and P_e is the electrical output power which depends on the terminal voltage V, the generator excitation voltage E and the reactance between the two buses. Equation (3) can also be written as [6]

$$M\frac{d^2X}{dt^2} = (P_m - P_e) \qquad (5)$$

Where

$$M = \frac{H}{\pi f} \qquad (6)$$

Equation (5) governs the generator voltage angles and hence it is called 'the swing equation'. In this work it is assumed that the damping of the generator is proportional to the generator's own relative speed or slip. It is often desirable to include a component of damping not accounted for in the calculation of P_e separately. This is accomplished by adding a term proportional to speed deviation in the above equation as follows [7]:

$$P_D = D\frac{dX}{dt} \qquad (7)$$

where D is the damping constant. Using the above equations, the original equation (1) may be written as follows;

$$M\frac{d^2X}{dt^2} = P_m - P_e - P_D \qquad (8)$$

The inertia constant (M) for each of the generators in the 14-bus system from the given data in Table A1 is evaluated at the synchronous speed and is considered to remain constant and it may be calculated as follows [8]:

$$M = \frac{GH}{S_{bas} \times 180 f} \qquad (9)$$

$$M_1 = \frac{615 \times 5.148}{100 \times 180 \times 50}$$

$$= 3.52 \times 10^{-3}$$

$$M_2 = M_3 = \frac{60 \times 6.54}{100 \times 180 \times 50}$$

$$= 0.436 \times 10^{-3}$$

$$M_4 = M_5 = \frac{25 \times 5.06}{100 \times 180 \times 50}$$

$$= 1.406 \times 10^{-4}$$

Where M is the inertia constant in p.u, G is the generating station rating, S_{bas} is the base rating MVA and f is the frequency which is 50Hz. The subtransient reactance in p.u. for each generating station on 100MVA base is given in Appendix A.

2.2 IEEE 14-bus Model

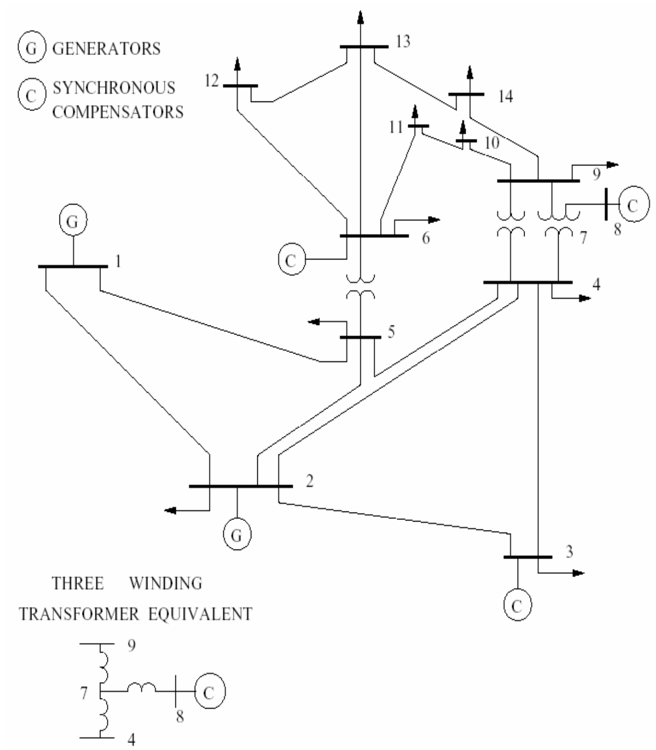

Fig. 1. IEEE 14BUS system [1]

The system that is studied is the IEEE 14-bus network (Figure 1). The stability problem of such a power system concerns the transmission of the power from one group of synchronous machines to another. During disturbances, the machines of each group swing more of less together, that is they retain approximately their relative angular position although these vary greatly with respect to the machines of the other group. For purposes of analysis the machines of each group can be replaced by one equivalent machine [10].

To perform stability study on the power system, the system is reduced using Newton-Raphson method to find the equivalent bus admittance. The elements of the admittance matrix Y represent the connections among the buses including those of the shunt, load and transmission lines as admittances. The nondiagonal elements, in the connection matrix Y are the admittances that connect the corresponding buses after being multiplied by -1, and the diagonal elements are summation of the equivalent load admittances corresponding to the buses and admittances and the shunt admittances of the lines that are connected to the corresponding buses. The internal bus admittance is obtained by applying the reduction method as follows [6]:

Ohm's Law

$$\begin{bmatrix} Y_{nn} & Y_{nm} \\ Y_{mn} & Y_{mm} \end{bmatrix}\begin{bmatrix} V \\ E \end{bmatrix}=\begin{bmatrix} 0 \\ I \end{bmatrix} \tag{10}$$

where

$$Y=\begin{bmatrix} Y_{nn} & Y_{nm} \\ Y_{mn} & Y_{mm} \end{bmatrix} \tag{11}$$

Y_{nn} is n × n matrix where n is the number of generators. Y_{nn} is m × m matrix where m is the number of buses that are connected to any generator. Y_{mn} is m × n matrix and Y_{nm} is n × m matrix.

$$I=[I_1 \quad \cdots \quad I_m] \tag{12}$$

I is the m vector of machine currents

$$I=[V_1 \quad \cdots \quad V_n] \tag{13}$$

V is the n vector of bus currents

$$I=[E_1 \quad \cdots \quad E_m] \tag{14}$$

The entire augmented matrix of the IEEE 14 test bus system is displayed in Matrix B1.

Computing synchronous-machines angles and frequency changes trajectories require the solution of the swing equations which may be written as follows:

$$M_i \frac{d^2 X(i)}{dt^2} = P_{mi} - E_i^2 R_{ii}$$

$$-\sum_{\substack{j=1 \\ j\neq i}}^{j=3} C_{ij} \sin\left[X(i)-X(j)+(90^\circ-\theta_{ij})\right] \tag{17}$$

$$-D_i \frac{dX(i)}{dt}$$

where

i = 1, 2,3,4,5 is the number of synchronous machine in the IEEE 14 bus system.

$C_{ij} = |E_i||E_j||Y_{ij}|$ is the power transferred at bus ij

$E_i = |E_i|\angle[X^e(i)]$ is the voltage at bus i.

$X^e(i)$ is the transfer reactance angle at bus i.

$Y_{ij} = |Y_{ij}| < \theta$ are the internal elements of matrix Y. R_{ii} are the real values of the diagonal elements of Y.

3. TRANSIENT STABILITY ANALYSIS OF THE IEEE 14-BUS SYSTEM

Power system faults may be categorized as one of four types in according to their effect on system stability: single line to ground, line to line, double line to ground and balanced three - phase. The first three types constitute severe unbalanced operating conditions. The fourth type is the most severe switching action with regard to the transient stability and it is selected to be investigated in the paper. Therefore a three phase fault is considered at two locations in Fig 1. One of them is on line 2-4 to the generating station 2 which has the smallest inertia value. The second is on line 10-11, very close to bus 10 and far away from all the generating station. Thus, the effect of the distance between the fault location and the generating stations and the effect of fault clearing time is analyzed.

The steps that being followed when doing the investigation are as follows:

- After modeling of the network, the computation of initial bus voltages, initial machine input and power output took place.
- Compute the internal machine voltages
- Compute the prefault system admittance matrix ($Y_{prefault}$)
- Set $t = 0$, set the faulted bus voltages to zero
- Compute the faulted system admittance matrix (Y_{fault})
- Isolate the line on which the fault occurred
- Compute the postfault system admittance matrix ($Y_{postfault}$)
- Plot the angular deviation for machines 2-5

The above steps were carried out with programms developed in the environment of MATLAB/SIMULINK and the Power System Toolbox (PST) package. The results are given below:

3.1 Fault on line 2-4

This study is performed with the intention of analyzing the effect of fault location in conjunction with the fault clearing time.

Fig. 2 shows the results of the angle differences of the machines in the system. Two faults located on two different lines are considered one closer to the generating stations and the other one far from the generating stations.

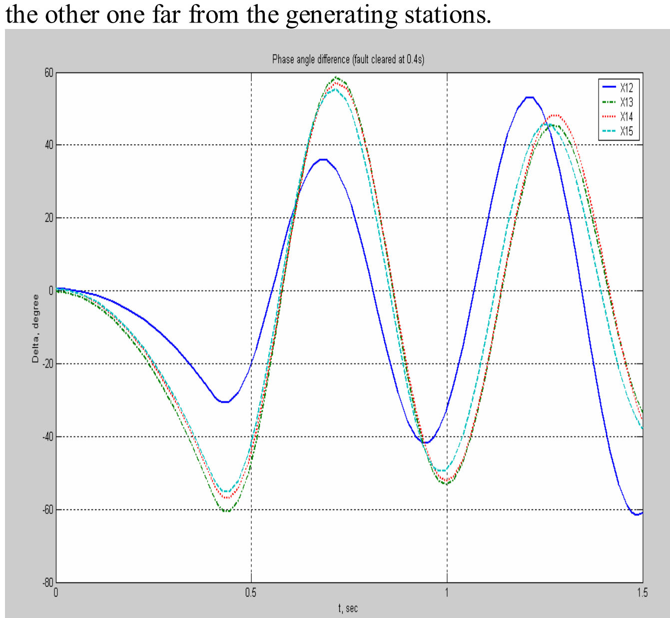

Fig. 2. Plots of angle differences for machines 2, 3, 4&5. Fault on bus 2 fault cleared at 0.4sec

The method which is applied to calculate the matrices Y_{fault} and $Y_{postfault}$ which describe the system during the fault and after clearing the fault, respectively, is the same as that applied to calculate $Y_{prefault}$. However, to find Y_{fault}, all the elements of the row and column which correspond to the faulted bus should be replaced in $Y_{prefault}$ by zeros and then the reduction method is applied. To find $Y_{postfault}$ the faulted line is isolated first by replacing the admittance of the faulted line by zero and calculating the diagonal elements of the buses connected by the faulted line before the fault occurs. Suppose a fault has occurred on line 2-4 at a point very close to bus 2, faulted reduced bus admittance matrix (Y_{fault}) and postfault reduced bus admittance matrix $Y_{postfault}$ are given by Matrix B2 and Matrix B3 respectively.

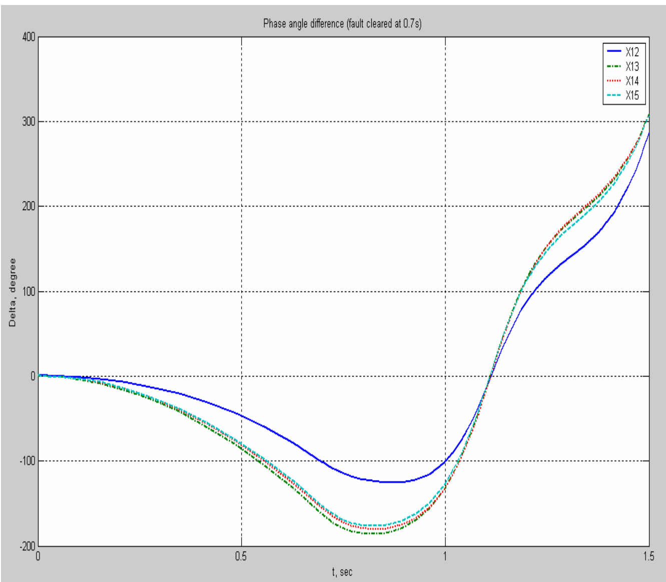

Fig. 3. Plots of angle differences for machines 2, 3, 4 & 5. Fault on bus 2 fault cleared at 0.6sec.

3.2 Fault on line 13-14

The fault occurs on line 10-11 very close to bus 10 and far away from the generating stations. The faulted reduced bus admittance matrix (Y_{fault}) is presented in Matrix B4.

After fault clearing by isolating the line between bus 13 and 14 the connection postfault reduced bus admittance matrix ($Y_{postfault}$) is shown in Matrix B5 and the results are presented on Fig.4 and Fig.5 respectively.

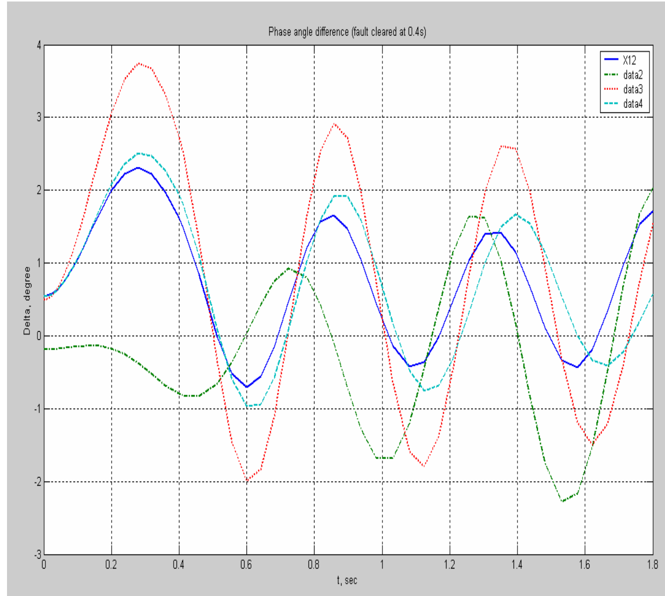

Fig. 4. plots of angle differences for machines 2, 3, 4 & 5.Fault on bus 10 fault cleared at 0.4sec

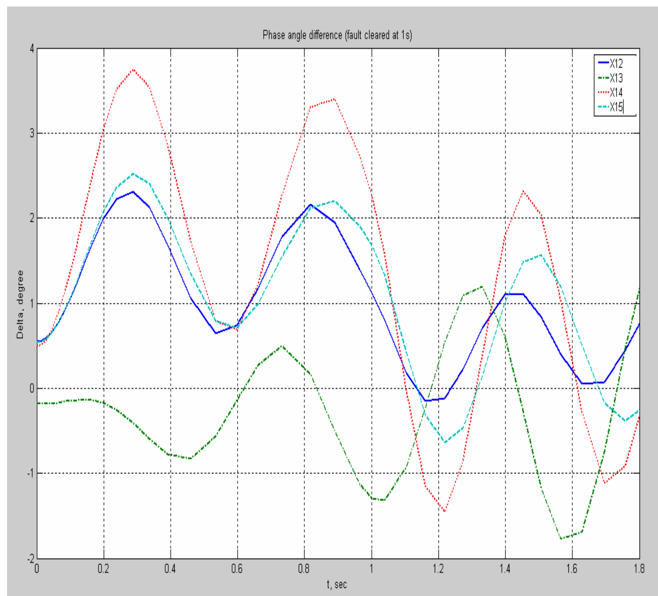

Fig. 5. plots of angle differences for machines 2, 3, 4 & 5. Fault on bus 10 fault cleared at 1sec.

4. DISCUSSION OF THE EFFECT OF FAULT LOCATION AND FAULT CLEARING TIME

There are many factors affecting the critical clearing time. Here, the effect distance between the fault location and the generating stations is studied.

Two fault locations for the same values of the damping and inertia constants are considered. One of the faults is on line (2-4), very close to bus 2 which is connected to machine 2, and the second fault is on line (10,11), very close to bus 10 .

Machines 4 and 5 have the smallest value of inertia, so they are expected to go out of step first.

In Fig.2. it may be seen that X12, X13, X14 and X15 increases during time (0.2s), where 0.4s is the clearing time of the fault. However X12, X13, X14 and X15 swing together and the solutions are carried out for X12, X13, X14 or X15. From this it can be said that the system is stable. Fig.3 shows the angular positions of the machines for a clearing time not equal to the critical clearing time($t_c \neq t_{cr}$) hence greater than the critical clearing time. In this case critical clearing time is =0.5s. If the clearing time increases beyond the critical clearing time the machines will go out of step as can be seen in Fig 3 whereby *tc>tcr*. In this case *tc = 0.6s*, after the fault is cleared at *tc = 0.6s* the *X12, X13, X14* and *X15* continued to decrease. From this it can be said that the system is unstable.

Fig 5 shows the swing curve for a fault on line (10-11). It is found that the critical clearing time *tcr* for the fault on line 10 is 0.6s, which is higher than the critical clearing time of the fault on line (1-4). This can be interpreted in such a way that the fault that is closer to the generating station must be cleared rapidly than the fault on the line far from the generation station.

Rapid clearing of the faults promotes power system stability. Indeed, increasing the speed of fault clearing is often the most effective and economical way to improve the transient stability of the power system. Rapid fault clearing is desirable for an additional reason. It decreases the damage to overhead transmission line conductors and insulators at the point of fault. After a fault occurs the synchronous machines' rotors will accelerate differently, so the angle between two equivalent machines will not be constant, and the system may become unstable. To determine whether a power system is stable after disturbances, it is necessary, in general, to plot and inspect the swing curves. If these curves show that the angle between any two equivalent machines tends to increase without limit, the system is unstable. If, on the other hand, after all disturbances, the angle between the two equivalent machines of every possible pair reaches maximum values and thereafter decrease, it is probable, although not certain, that the system is stable.

5. Conclusion

In this paper the stability of the IEEE 14-bus system has been studied. The stability has determined by plotting the swing equations which are nonlinear. For a three-phase fault at two different locations in the system, the reduction method has been applied to determine the reduced connection matrices Y_{fault} and $Y_{postfault}$ during and after clearing the fault. The two different locations of the fault are chosen to observe the effect of the distance of the fault from the generators on the stability. It is therefore recommended that power system designers must do proper inspections on the design with regard to transient stability as it is the most cumbersome aspect in power system design.

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Appendix A

Table A.1 Generator Data [1]

Generator bus no.	1	2	3	4	5
MVA	615	60	60	25	25
x_1 (p.u.)	0.24	0	0	0.134	0.134
r_a (p.u.)	0	0	0.003	0.001	0.004
x_d (p.u.)	0.898	1.05	1.05	1.25	1.25
x''_d (p.u.)	0.3	0.19	0.185	0.232	0.232
x'''_d(p.u.)	0.23	0.13	0.13	0.12	0.12
T'_{do}	7.4	6.1	6.1	4.75	4.75
T''_{do}	0.03	0.04	0.04	0.06	0.06
x_q (p.u.)	0.646	0.98	0.98	1.22	1.22
x'_q (p.u.)	0.646	0.36	0.36	0.715	0.715
x''_q (p.u.)	0.4	0.13	0.13	0.12	0.12
T'_{qo}	0	0.3	0.3	1.5	1.5
T'_{qo}	0.033	0.1	0.099	0.21	0.21
H	5.148	6.54	6.54	5.06	5.06
D	2	2	2	2	2

Table A2. Line Data [1]

Bus	P Gen. (pu)	Q Gen. pu	P Load (pu)	Q Load (pu)	Bus Type	Q Gen. max (pu)	Q Gen. min. (pu)
1	2.32	0	0	0	2	10	-10
2	0.4	-0.424	0.217	0.127	1	0.5	-0.4
3	0	0	0.942	0.19	2	0.4	0
4	0	0	0.478	0	3	0	0
5	0	0	0.076	0.016	3	0	0
6	0	0	0.112	0.075	2	0.24	-0.06
7	0	0	0	0	3	0	0
8	0	0	0	0	2	0.24	-0.06
9	0	0	0.295	0.166	3	0	0
10	0	0	0.09	0.058	3	0	0
11	0	0	0.035	0.018	3	0	0
12	0	0	0.061	0.016	3	0	0
13	0	0	0.135	0.058	3	0	0
14	0	0	0.149	0.05	3	0	0

*Bus Type: (1) swing bus, (2) generator bus (PV bus), and (3) load bus (PQ bus)

Table A.3. Line Data [1]

From	To	R(pu)	X(pu)	B(pu)	Tap

Bus	Bus				Ratio
1	2	0.0194	0.05917	0.053	1
1	5	0.054	0.22304	0.049	1
2	3	0.047	0.19797	0.044	1
2	4	0.0581	0.17632	0.037	1
2	5	0.057	0.17388	0.034	1
3	4	0.067	0.17103	0.346	1
4	5	0.0134	0.04211	0.013	1
4	7	0	0.20912	0	1
4	9	0	0.55618	0	1
5	6	0	0.25202	0	0.9
6	11	0.095	0.1989	0	1
6	12	0.1229	0.25581	0	1
6	13	0.0662	0.13027	0	1
7	8	0	0.17615	0	1
7	9	0	0.11001	0	1
9	10	0.0318	0.0845	0	1
9	14	0.1271	0.27038	0	1
10	11	0.0821	0.19207	0	1
12	13	0.2209	0.19988	0	1
13	14	0.1709	0.34802	0	1

APPENDIX B

Matrix B1. The equivalent admittance prefault Matrix ($Y_{prefault}$) for the IEEE 14-test bus system.

$$\begin{bmatrix} 0.1017 - 2.4199i & 0.0096 + 0.9097i & -0.0567 + 0.5006i & -0.0322 + 0.4973i & -0.0186 + 0.5531i \\ 0.0096 + 0.9097i & 0.1375 - 3.5521i & -0.0574 + 0.9568i & -0.0437 + 0.8608i & -0.0395 + 0.8903i \\ -0.0567 + 0.5006i & -0.0574 + 0.9568i & 0.2630 - 2.9134i & -0.0640 + 0.8148i & -0.0791 + 0.6968i \\ -0.0322 + 0.4973i & -0.0437 + 0.8608i & -0.0640 + 0.8148i & 0.1156 - 3.0762i & 0.0303 + 0.9547i \\ -0.0186 + 0.5531i & -0.0395 + 0.8903i & -0.0791 + 0.6968i & 0.0303 + 0.9547i & 0.1117 - 3.0513i \end{bmatrix}$$

Matrix B2. The equivalent admittance fault Matrix (Y_{fault}) for the IEEE 14-test bus system with the fault nearby bus 2.

$$\begin{bmatrix} 0.1017 - 2.3981i & 0.0138 + 0.9699i & -0.0540 + 0.4950i & -0.0374 + 0.4411i & -0.0203 + 0.5276i \\ 0.0138 + 0.9699i & 0.1611 - 3.3869i & -0.0510 + 0.9417i & -0.0687 + 0.7091i & -0.0492 + 0.8216i \\ -0.0540 + 0.4950i & -0.0510 + 0.9417i & 0.2621 - 2.9140i & -0.0693 + 0.8252i & -0.0814 + 0.7007i \\ -0.0374 + 0.4411i & -0.0687 + 0.7091i & -0.0693 + 0.8252i & 0.1410 - 2.9451i & 0.0403 + 1.0124i \\ -0.0203 + 0.5276i & -0.0492 + 0.8216i & -0.0814 + 0.7007i & 0.0403 + 1.0124i & 0.1156 - 3.0263i \end{bmatrix}$$

Matrix B3. The equivalent admittance fault Matrix ($Y_{postfault}$) for the IEEE 14-test bus system when line 1-4 is removed.

$$\begin{bmatrix} 0.1018 - 2.4197i & 0.0096 + 0.9099i & -0.0568 + 0.5004i & -0.0325 + 0.4967i & -0.0182 + 0.5539i \\ 0.0096 + 0.9099i & 0.1375 - 3.5520i & -0.0574 + 0.9566i & -0.0439 + 0.8603i & -0.0393 + 0.8910i \\ -0.0568 + 0.5004i & -0.0574 + 0.9566i & 0.2631 - 2.9132i & -0.0637 + 0.8154i & -0.0795 + 0.6959i \\ -0.0325 + 0.4967i & -0.0439 + 0.8603i & -0.0637 + 0.8154i & 0.1169 - 3.0747i & 0.0288 + 0.9510i \\ -0.0182 + 0.5539i & -0.0393 + 0.8910i & -0.0795 + 0.6959i & 0.0288 + 0.9510i & 0.1136 - 3.0476i \end{bmatrix}$$

Matrix B4. The equivalent admittance fault Matrix (Y_{fault}) for the IEEE 14-test bus system with the fault nearby bus 10.

$$\begin{bmatrix} 0.1253 - 2.8646i & 0 & 0.0011 + 0.0350i & 0.0159 + 0.0781i & 0.0284 + 0.1194i \\ 0 & 0 - 5.4054i & 0 & 0 & 0 \\ 0.0011 + 0.0350i & 0 & 0.3583 - 3.3985i & 0.0176 + 0.3778i & 0.0024 + 0.2444i \\ 0.0159 + 0.0781i & 0 & 0.0176 + 0.3778i & 0.1855 - 3.4698i & 0.0999 + 0.5473i \\ 0.0284 + 0.1194i & 0 & 0.0024 + 0.2444i & 0.0999 + 0.5473i & 0.1809 - 3.4730i \end{bmatrix}$$

Matrix B5. The equivalent admittance fault Matrix ($Y_{postfault}$) for the IEEE 14-test bus system when line 10-11 is removed.

$$\begin{bmatrix} 0.1167 - 2.4776i & 0.0340 + 0.8110i & -0.0319 + 0.4148i & -0.0174 + 0.3670i & -0.0001 + 0.4319i \\ 0.0340 + 0.8110i & 0.1776 - 3.7165i & -0.0167 + 0.8119i & -0.0204 + 0.6413i & -0.0099 + 0.6860i \\ -0.0319 + 0.4148i & -0.0167 + 0.8119i & 0.3037 - 3.0410i & -0.0365 + 0.6203i & -0.0464 + 0.5161i \\ -0.0174 + 0.3670i & -0.0204 + 0.6413i & -0.0365 + 0.6203i & 0.1077 - 3.3643i & 0.0332 + 0.6854i \\ -0.0001 + 0.4319i & -0.0099 + 0.6860i & -0.0464 + 0.5161i & 0.0332 + 0.6854i & 0.1240 - 3.3027i \end{bmatrix}$$

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